
External Menisci on Thin Inclined Fibers with Circular Cross-section (the Case of Full Wetting)

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Abstract—An analytical solution is constructed for the problem of determining the configuration of external menisci that forms on the surface of a liquid when a thin inclined fiber with circular cross-section is immersed in this liquid. It is assumed that the liquid completely wets the fiber. We found that the fiber inclination leads to an asymmetry of both the contact line and the meniscus as a whole. It is shown that the maximum of the liquid rise increases with increasing angle of inclination of the fiber from the vertical. At the same time, some part of the meniscus falls even below the horizontal level of the initially undisturbed liquid.

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INTRODUCTION

A single fiber, being immersed in a liquid, causes its capillary rise. This leads to a change in the shape of the initially undisturbed liquid surface and, accordingly, to the formation of some external meniscus. Its equilibrium configuration strongly depends on both the wetting properties of fibers and the shape of fibers cross-section [1]. The analysis of external menisci on vertically immersed fiber was carried out in the papers [2]–[4]. The case when the fiber is thin and it becomes possible to use the asymptotic approach is the most studied [5]–[8]. At the same time, such an important factor as fiber inclination was practically not studied. In the paper [9], the asymptotic approach [5] was extended to the case when the thin fiber is immersed in a liquid so that the fiber axis is inclined to the vertical at an arbitrary angle. As a result, an asymptotic formulation for the problem of determining the configuration of external menisci in this case was obtained. Our purpose is to construct an example of using the new asymptotic formulation for the case of thin inclined fibers [9].

1. THE PROBLEM

Let a circular fiber be immersed in a liquid so that the central axis of the fiber lies in the vertical plane xz , and the angle between the fiber axis and the vertical axis z is equal to $\alpha_0 \in (0, \pi/2)$, see Fig. 1 a. The horizontal plane xy is actually the surface of the initially undisturbed liquid. The material of the fiber is homogeneous, and the fiber wettability is characterized by a contact angle γ_0 . Due to the capillary rise of the liquid on the fiber, the liquid surface is deformed. As a result, some external meniscus Σ is established. Also, some contact line Γ_c is established at the boundary of the three media “air – liquid – fiber”. We want to find the configuration of the external meniscus Σ . Due to the mirror symmetry relative to the vertical xz -plane, it is sufficient to find this configuration only in the half-space $y \geq 0$.

We choose the radius L_m of the circular cross-section of the fiber as the characteristic linear scale. Then, the Cartesian coordinate system xyz in Fig. 1 a can be considered as a dimensionless system. Similar to the case of a vertical fiber [5], in this problem, there is another linear scale,

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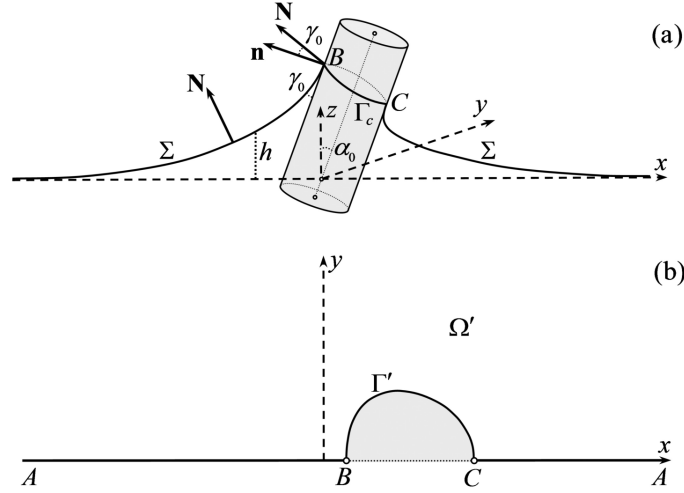


Fig. 1. (a) Schematic representation of the external meniscus Σ on an inclined fiber with circular cross-section. (b) Projection of the symmetry element of Σ in half-space $y \geq 0$ onto the xy -plane. Vector $\mathbf{N} = (N_x, N_y, N_z)$ denotes the normal to the meniscus Σ . Vector $\mathbf{n} = (n_x, n_y, n_z)$ denotes the normal to the lateral surface of the fiber.

namely the capillary length $L_c = \sqrt{\sigma/(\rho g)}$, where σ is the surface tension, ρ is the density of the liquid, and g is the gravitational acceleration. Then, the resulting effect of capillary rise of liquid on the fiber will be characterized by the dimensionless Bond number ε [10]

$$\varepsilon = \frac{L_m^2}{L_c^2} = \frac{\rho g L_m^2}{\sigma}.$$

The symmetry element of the meniscus Σ in the half-space $y > 0$ can be defined by a function $h(x, y)$

$$\Omega' : \quad z = h(x, y), \quad (1)$$

where the value of $h(x, y)$ is the dimensionless height of the meniscus, which is measured from level $z = 0$, see Fig. 1 a; Ω' is a region obtained by the projection of the symmetry element of the meniscus Σ onto the horizontal plane xy , see Fig. 1 b. The region Ω' is limited by the contour Γ' that is the projection of the contact line Γ_c on the xy -plane. Since the shape of the contact line Γ_c is unknown, the shape of the contour Γ' is also unknown.

Using function $h(x, y)$, we can find an analytical expression for the normal $\mathbf{N}$ to the meniscus Σ [11]

$$\mathbf{N} = (1 + |\nabla h|^2)^{-1/2} \left(-\frac{\partial h}{\partial x}, -\frac{\partial h}{\partial y}, 1 \right). \quad (2)$$

Then, the mathematical model of the capillary rise of the liquid on the inclined fiber can be formally written in a form similar to that of the vertical fiber [5]

$$\Omega' : \quad \nabla \cdot \left[(1 + |\nabla h|^2)^{-1/2} \nabla h \right] - \varepsilon h = 0, \quad (3)$$

$$\Gamma' : \quad \mathbf{N} \cdot \mathbf{n} = \cos \gamma_0, \quad (4)$$

$$AB \cup CA : \quad \partial h / \partial y = 0, \quad (5)$$

$$r = \sqrt{x^2 + y^2} \rightarrow \infty : \quad h(x, y) \rightarrow 0. \quad (6)$$

Here the first term of equation (3) represents the average curvature of the meniscus Σ [11], the second term is the hydrostatic pressure, and the equation as a whole is Laplace's capillary law [12]. Vector $\mathbf{n} = (n_x, n_y, n_z)$ denotes the normal to the lateral surface of the fiber. The boundary condition (4) is the Young-Laplace condition of dynamic equilibrium of the contact line Γ_c [1]. The boundary condition (5) is the condition of smooth symmetrical continuation of the meniscus Σ across the boundaries AB and CA , see Fig. 1 a. If the fiber cross-section is given, then the boundary value problem (3)–(6) depends on the two physical parameters, namely on the contact angle γ_0 and on the angle of fiber inclination α_0 .

Let us emphasize two features of this problem that significantly complicate its analysis in comparison with the case of the vertical fiber:

(i) Even in the simplest case of the inclined fiber with the circular cross-section, the condition (4) is written on the contour Γ' of unknown shape. The shape of Γ' must also be determined.

(ii) The use of the function (1) is possible only under the restriction $\gamma_0 \geq \alpha_0$ since its violation leads to loss of one-valuedness for the function (1) in the vicinity of the point C , see Fig. 1a. At the same time the all physical assumptions concerning the problem (3)–(6) remain valid.

2. ASYMPTOTIC APPROACH FOR THIN FIBERS

Let the fiber be thin, i.e., $L_m \ll L_c$. Then, the Bond number becomes small, $\varepsilon \ll 1$, and asymptotic methods can be used. The well-known asymptotic approach for the case of the vertical thin fiber [5] allows a formal generalization to the case of the inclined thin fiber; but such a simple formal generalization is ineffective, since both indicated features (i) and (ii) of problem (3)–(6) are preserved for small $\varepsilon \ll 1$, see [9]. For example, even the simplest case of the inclined thin fiber with the circular cross-section cannot be analyzed. At the same time, in the article [9] it was shown that the approach [5] can be supplemented by a transformation to the new Cartesian coordinate system that is associated with the inclined fiber. This allows eliminating both indicated features (i) and (ii) of the problem (3)–(6).

Now, following [9], we introduce a new Cartesian coordinate system XYZ , which is obtained by rotating the coordinate system xyz around the axis y by the angle of α_0 , see Fig. 2 a. Then, the axis Z coincides with the central axis of the fiber and the plane XY is strictly perpendicular to the fiber axis. Obviously, we have identity

$$Y = y, \quad (7)$$

and, therefore, we are dealing with a simpler transformation $xz \rightarrow XZ$ of the Cartesian coordinate system xz by rotating it relative to the origin by the angle $(-\alpha_0) < 0$. Here we use the generally accepted rule in analytical geometry for choosing positive directions of angles [13].

Let us present the resulting formulas for both the direct transformation of coordinates $xz \rightarrow XZ$ and the inverse transformation of coordinates $XZ \rightarrow xz$ [13]

$$\begin{cases} X = x \cos \alpha_0 - z \sin \alpha_0, \\ Z = x \sin \alpha_0 + z \cos \alpha_0, \end{cases} \Leftrightarrow \begin{cases} x = X \cos \alpha_0 + Z \sin \alpha_0, \\ z = -X \sin \alpha_0 + Z \cos \alpha_0. \end{cases} \quad (8)$$

Now we can make the transition from the definition of the surface Σ by using the function $h(x, y)$ in space (x, y, z) , see (1), to the definition of the same surface by using a function $H(X, Y)$ in space (X, Y, Z)

$$\Omega : \quad Z = H(X, Y), \quad (9)$$

where Ω is a region obtained by the projection of the symmetry element of the surface Σ in the half-space $Y > 0$ on the plane XY , see Fig. 2 b; the value of $H(X, Y)$ is the Z -coordinate of the meniscus Σ at the point $(X, Y) \in \Omega$, see Fig. 2 a. We note that, in contrast to the region Ω' , Ω is the region of known shape. Indeed, it is clear from Fig. 2 b that region Ω is the upper half of the exterior of the contour Γ (unit circle). Note also that the function (9) can be used for any combination of the angles γ_0 and α_0 , since $H(X, Y)$ always remains the one-valued function for typical configurations of the meniscus Σ , see Fig. 2 b. Thus, having made the transition from the

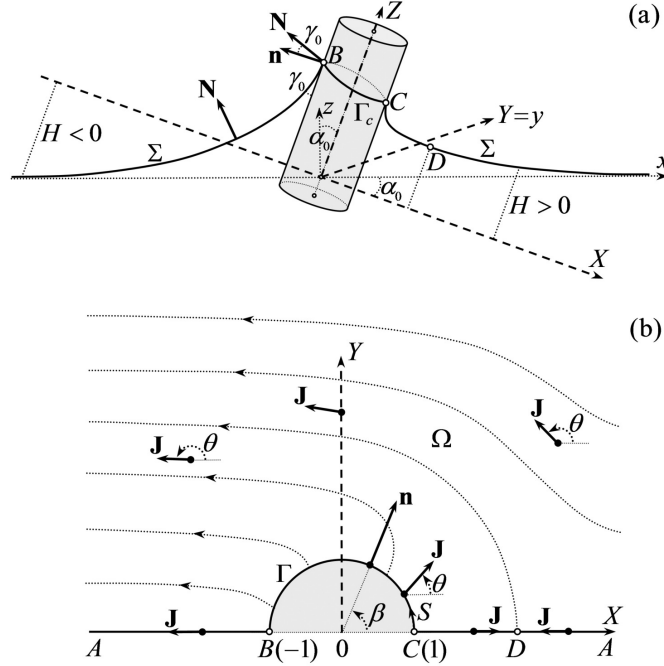


Fig. 2. (a) Schematic representation of the external meniscus Σ in the Cartesian coordinate system XYZ associated with the fiber. (b) Projection of the symmetry element of Σ in half-space $Y \geq 0$ onto the XY -plane; the dashed lines show the streamline pattern for the fictitious flow $\mathbf{J}$.

function $h(x, y)$ to the function $H(X, Y)$, we got rid of both indicated features (i) and (ii) of the problem (3)–(6).

Taking into account the formula (8), we can establish a relation between the functions $h(x, y)$ and $H(X, Y)$

$$h(x, y) = -X \sin \alpha_0 + H(X, Y) \cos \alpha_0.$$

In particular, it follows from this relation that there are two equivalent records for an equation of a horizontal plane as some approximate equation of the meniscus Σ at infinity

$$r \rightarrow \infty : \quad z \approx 0 \quad \Leftrightarrow \quad R \rightarrow \infty : \quad Z \approx X \tan \alpha_0, \quad (10)$$

where $r = \sqrt{x^2 + y^2}$ and $R = \sqrt{X^2 + Y^2}$.

The normal $\mathbf{N}$ to the meniscus Σ can be expressed in terms of the function $H(X, Y)$ by analogy with (2)

$$\mathbf{N} = (1 + |\nabla H(X, Y)|^2)^{-1/2} \left(-\frac{\partial H}{\partial X}, -\frac{\partial H}{\partial Y}, 1 \right). \quad (11)$$

Then, the problem (3)–(6) can be rewritten in terms of the function $H(X, Y)$, next the assumption $\varepsilon \ll 1$ can be accepted, and the method of asymptotic matching can be applied to this rewritten problem. This was done in the paper [9]. As a result, it was shown that the outer asymptotic expansion $H \approx H^{(o)}(X, Y)$ approximates the meniscus Σ in the zone $R \sim \varepsilon^{-1/2}$. The leading term of this expansion has the form [9]

$$H^{(o)}(X, Y) = X \tan \alpha_0 + \frac{Q}{2\pi \cos \alpha_0} K_0 \left(\varepsilon^{1/2} \sqrt{\frac{X^2}{\cos^2 \alpha_0} + Y^2} \right), \quad (12)$$

where K_0 is the Bessel function of the second kind [14] and Q is a parameter of the asymptotic matching (an undetermined real constant).

The inner asymptotic expansion $H \approx H^{(i)}(X, Y)$ approximates the meniscus Σ in the fiber neighborhood $R \sim 1$. The leading term of this expansion satisfies the boundary value problem [9]

$$\Omega : \quad \nabla \cdot \left[\left(1 + |\nabla H^{(i)}|^2 \right)^{-1/2} \nabla H^{(i)} \right] = 0, \quad (13)$$

$$\Gamma : \quad - \left(1 + |\nabla H^{(i)}|^2 \right)^{-1/2} \nabla H^{(i)} \cdot \mathbf{n} = \cos \gamma_0, \quad (14)$$

$$AB \cup CA : \quad \partial H^{(i)} / \partial Y = 0, \quad (15)$$

$$R \gg 1 : \quad H^{(i)}(X, Y) = X \tan \alpha_0 - \frac{Q}{2\pi \cos \alpha_0} \left[\ln \sqrt{\frac{X^2}{\cos^2 \alpha_0} + Y^2} + C_\varepsilon \right] + O(R^{-1} \ln R). \quad (16)$$

Here we denote as C_ε such constant

$$C_\varepsilon = \ln \left(0.5 \varepsilon^{1/2} e^E \right), \quad (17)$$

where $E = 0.57\dots$ is the Euler constant [14].

The governing differential equation (13) is essentially the minimal surface equation, as in the case of the vertical thin fiber [5]. The condition at infinity (16) is the result of matching with the main term of the outer asymptotic expansion (12). The Young–Laplace condition (14) essentially coincides with the condition (4). The vector $\mathbf{n}$ also denotes the normal to the fiber surface. However, in the new coordinate system XYZ the Z -component of this normal is equal to zero. If we introduce the notation β for the inclination angle of the normal $\mathbf{n}$ to the X -axis, then we get an analytical expression: $\mathbf{n} = (\cos \beta, \sin \beta, 0)$ (see Fig. 2b).

Suppose that the solution of the problem (13)–(16) has been found, i.e., that the leading term of the inner asymptotic expansion $H^{(i)}(X, Y)$ is known. Then, using the procedure of the asymptotic matching [15], we can find the leading term of the asymptotic expansion $H^{(u)}(X, Y)$, which uniformly approximates the meniscus Σ everywhere in the XY -plane [9]

$$H^{(u)}(X, Y) = H^{(i)}(X, Y) + \frac{Q}{2\pi \cos \alpha_0} \left[K_0 \left(\varepsilon^{1/2} \sqrt{\frac{X^2}{\cos^2 \alpha_0} + Y^2} \right) + \ln \sqrt{\frac{X^2}{\cos^2 \alpha_0} + Y^2} + C_\varepsilon \right]. \quad (18)$$

In addition, the parameter Q can be found in advance, even before solving the problem (13)–(16), by using the formula [9]

$$Q = \frac{l \cos \gamma_0}{\cos \alpha_0} = \frac{2\pi \cos \gamma_0}{\cos \alpha_0}, \quad (19)$$

since we know a priori the perimeter l of the fiber cross-section Γ (unit circle): $l = 2\pi$.

It only remains to find the solution $H^{(i)}(X, Y)$ to the problem of the internal asymptotic expansion (13)–(16). Note that the behavior of the function $H^{(i)}(X, Y)$ at infinity in this problem differs from the case of the vertical thin fiber [5]. In addition to the logarithmic singularity, here we have a higher order singularity, namely the linear term $X \tan \alpha_0$, see the condition (16). The structure of the logarithmic term itself is also slightly different from [5]. Nevertheless, we can hope that some solutions of the problem (13)–(16) can be constructed if we apply the same mathematical apparatus that was effectively applied in the case of the vertical thin fiber. This apparatus includes an analogy with the theory of flows of non-Newtonian fluids through porous media [16] and the Chaplygin's hodograph transformation [17].

3. ANALOGY WITH FLOWS OF NON-NEWTONIAN FLUIDS THROUGH POROUS MEDIA. CHAPLYGIN'S HODOGRAPH TRANSFORMATION.

Let us define a fictitious flow of incompressible fluid through a porous medium $\mathbf{J}$ such that

$$\Omega: \quad \mathbf{J} = -\frac{\nabla H^{(i)}}{\sqrt{1+|\nabla H^{(i)}|^2}}, \quad \nabla \cdot \mathbf{J} = 0, \quad (20)$$

where the function $H^{(i)}(X, Y)$ is an analogue of the pressure in the liquid [16]. Here the first relation represents the nonlinear law of flow of non-Newtonian fluid in porous media, the second relation is the continuity equation, and the formula (20) as a whole represents another form of writing for the equation (13).

The fictitious flow $\mathbf{J}$ is characterized by its magnitude J and by its inclination angle θ to the X -axis, i.e. $\mathbf{J} = J(\cos \theta, \sin \theta)$. If we compare the formulas (20) and (11), we come to the conclusion that the fictitious flow $\mathbf{J}$ is equal to the projection of the normal $\mathbf{N}$ onto the XY -plane. Since we have some general representation of the shape of the meniscus Σ , we can imagine the behavior of the vector field $\mathbf{J}(X, Y)$ in region Ω . As a result, we can imagine the streamline pattern that corresponds to the fictitious flow $\mathbf{J}$, see dotted lines in Fig. 2 b. In particular, it is easy to establish that at some point $D \in CA$ the tangent to the meniscus Σ is strictly parallel to the XY -plane. Accordingly, for the vector field $\mathbf{J}(X, Y)$ of the fictitious flow, the point D is the flow stagnation point for which $J = 0$.

Since the fluid is incompressible, we can introduce a stream function $\psi(X, Y)$ [18]. Using the formula (20), we obtain

$$J = |\nabla H^{(i)}| (1 + |\nabla H^{(i)}|^2)^{-1/2} \Leftrightarrow |\nabla H^{(i)}| = J(1 - J^2)^{-1/2}.$$

This one-to-one correspondence between the values J and $|\nabla H^{(i)}|$ allows us to use of the Chaplygin's hodograph transformation [17]. All the mathematical details of the application of this transformation to the problems of capillary rise of liquid along thin fibers have already been published many times, see [5]–[8]. Therefore, here we present only a summary of the main formulas for applying the Chaplygin's hodograph transformation to the problem (13)–(16):

$$J_X = J \cos \theta = \frac{\partial \psi}{\partial Y}, \quad J_Y = J \sin \theta = -\frac{\partial \psi}{\partial X}, \quad (21)$$

$$\nabla H^{(i)}(X, Y) = -(1 - J^2)^{-1/2} \mathbf{J},$$

$$J = \frac{1}{\cosh t} \Leftrightarrow t = \operatorname{arccosh} \frac{1}{J}, \quad (22)$$

$$0 \leq |\nabla H^{(i)}(X, Y)| \leq \infty, \quad 0 \leq J \leq 1, \quad 0 \leq t \leq \infty,$$

$$\frac{\partial \psi}{\partial t} = \frac{\partial H^{(i)}}{\partial \theta}, \quad \frac{\partial \psi}{\partial \theta} = -\frac{\partial H^{(i)}}{\partial t}. \quad (23)$$

As a result of applying the Chaplygin's transformation, we arrive at the Cauchy–Riemann equations (23), which allows us to introduce complex variables W and χ

$$W = -H^{(i)} + i\psi, \quad \chi = t + i\theta,$$

where W is called the complex velocity potential, and χ is called the hodograph (these are conventional names, see [16]). Accordingly, we can use two functions of the complex variables, namely $W(\chi)$ and $\chi(W)$ [19].

If one of these functions is found, we can then return to the variables X and Y by writing Khristianovich's formulas for the total differentials of functions $X(H^{(i)}, \psi)$ and $Y(H^{(i)}, \psi)$ [16]

$$dX = -\cos \theta \sinh t dH^{(i)} - \sin \theta \cosh t d\psi, \quad dY = -\sin \theta \sinh t dH^{(i)} + \cos \theta \cosh t d\psi. \quad (24)$$

In order to find one of the functions $W(\chi)$ or $\chi(W)$, let us rewrite the boundary value problem (13)–(16) in terms of complex variables χ and W . Consider the differential equation (13). As shown above, a consequence of this equation is the fact that $W(\chi)$ and $\chi(W)$ are functions of complex variable.

Consider now the condition (14) at the boundary Γ . First, we write it in terms of the fictitious flow $\mathbf{J}$, using the formula (20)

$$\Gamma : \quad \mathbf{J} \cdot \mathbf{n} = \cos \gamma_0. \quad (25)$$

The scalar product on the left-hand side of (25) can be expressed through the difference between the angles θ and β : $\mathbf{J} \cdot \mathbf{n} = J \cos(\theta - \beta)$, see Fig. 2 b. Then, using the relationship (22) between the values J and t , the condition (25) can be rewritten in terms of the hodograph χ

$$\Gamma : \quad \cos(\theta - \beta) = \cos \gamma_0 \cosh t. \quad (26)$$

Note that this condition in the form (25) allows us to accurately determine the amount of fictitious fluid M entering per unit time the region Ω through the boundary Γ

$$M = \pi \cos \gamma_0. \quad (27)$$

Accordingly, the amount of fictitious fluid entering through the full cross section of the fiber is equal to $2M$. The fictitious fluid is incompressible, and the two remaining boundaries of the region Ω , namely AB and CA , are impermeable. Therefore, there must be a singularity at infinity, namely sink of the same strength $2M = 2\pi \cos \gamma_0$. In addition, according to both the condition (16) and the streamline pattern for the fictitious flow $\mathbf{J}$ in Fig. 2 b, there must be another singularity at infinity, namely the dipole singularity [18].

The condition (15) can easily be rewritten in terms of W and χ if we take into account the same streamline pattern in Fig. 2 b and the value of M , see (27)

$$CD \cup DA : \quad \psi = 0; \quad AB : \quad \psi = \pi \cos \gamma_0, \quad (28)$$

$$CD : \quad \theta = 0; \quad DA \cup AB : \quad \theta = \pi. \quad (29)$$

Finally, consider the condition (16) at infinity. The infinity $A \in DA$ corresponds to $X \rightarrow +\infty$ and the infinity $A \in AB$ corresponds to $X \rightarrow -\infty$, see Fig. 2b. It is clear that it is the first term in this condition that determines the behavior of the function $H^{(i)}(X, Y)$ at infinity. Taking into account the condition (28), we obtain directly from the condition (16)

$$A \in DA : \quad H^{(i)} \rightarrow +\infty, \quad \psi = 0; \quad A \in AB : \quad H^{(i)} \rightarrow -\infty, \quad \psi = \pi \cos \gamma_0. \quad (30)$$

The formula (30) expresses the condition at infinity (16) in terms of the complex velocity potential W . In order to find this condition in terms of the hodograph χ , it is necessary to obtain estimates of the fictitious flow $\mathbf{J}$ at infinity.

Let us introduce the following notation

$$\mathbf{E}_0 = (1, 0), \quad \mathbf{R} = (X, Y), \quad \mathbf{R}_1 = (X \cos^{-2} \alpha_0, Y).$$

Then, by differentiating the equation (16), we obtain the relation

$$R \gg 1 : \quad \nabla H^{(i)}(X, Y) = \tan \alpha_0 \mathbf{E}_0 - \left(\frac{Q}{2\pi \cos \alpha_0} \right) \frac{\mathbf{R}_1}{X^2 \cos^{-2} \alpha_0 + Y^2} + o(R^{-1}).$$

Substituting this relation into the first expression of (20), we obtain the estimate of the fictitious flow $\mathbf{J}$ at infinity (see [9] for details)

$$R \gg 1 : \quad \mathbf{J} = -\sin \alpha_0 \mathbf{E}_0 + \left(\frac{Q}{2\pi} \right) \frac{\mathbf{R}}{X^2 \cos^{-2} \alpha_0 + Y^2} + o(R^{-1}).$$

Using this, we can estimate the leading term of the asymptotic expansion of $\mathbf{J}$ in powers of R^{-1} as $\mathbf{J} \approx -\sin \alpha_0 \mathbf{E}_0$. As a result, the condition at infinity (16) in terms of the hodograph χ takes the

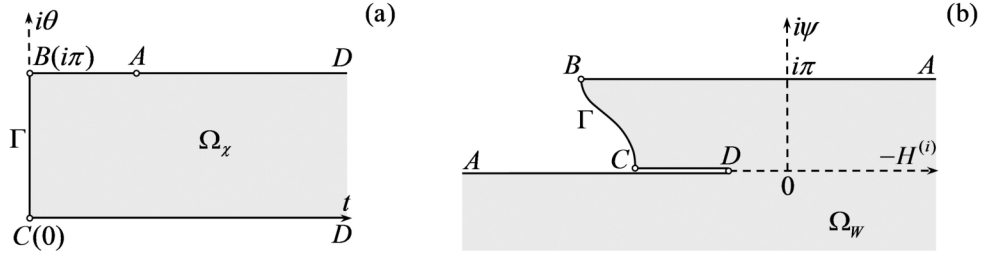


Fig. 3. (a) Plane of the hodograph $\chi = t + i\theta$. (b) Plane of the complex velocity potential $W = -H^{(i)} + i\psi$. In the W -plane, the shape of the boundary Γ is unknown.

form

$$A: \quad J_A = \sin \alpha_0, \quad t_A = \operatorname{arccosh} \left(\frac{1}{\sin \alpha_0} \right) = \ln \left(\frac{1 + \cos \alpha_0}{\sin \alpha_0} \right), \quad \theta_A = \pi, \quad (31)$$

where the formulas (22) and (29) were also taken into account.

4. THE PARTICULAR CASE OF FULL WETTING. THE χ - AND W -PLANES

Examples of determining the configuration of external menisci on vertical thin fibers [5], [6], [8] showed that the use of the Chaplygin's transformation is most effective in the particular case of full wetting of the fiber. We also limit ourselves to this particular case, i.e., we accept $\gamma_0 = 0$ here and in the sequel. Then, using the formula (19), we can immediately find the value of Q :

$$Q = \frac{2\pi}{\cos \alpha_0}. \quad (32)$$

It can easily be checked that in the particular case $\gamma_0 = 0$ the boundary conditions (28)–(30) take the form

$$CD \cup DA: \quad \psi = 0; \quad AB: \quad \psi = \pi, \quad (33)$$

$$CD: \quad \theta = 0; \quad DA \cup AB: \quad \theta = \pi, \quad (34)$$

$$A \in DA: \quad H^{(i)} \rightarrow +\infty, \quad \psi = 0; \quad A \in AB: \quad H^{(i)} \rightarrow -\infty, \quad \psi = \pi, \quad (35)$$

we must add to this system of equations the condition (31), the form of which does not change.

Let us try to determine the shape of the regions Ω_χ and Ω_W corresponding to the region Ω in the χ - and W -planes. For this purpose we use the boundary conditions in terms χ and W , which have been obtained in Section 3.

In the case of full wetting, at every point of the contact line Γ_c the normal to the meniscus $\mathbf{N}$ obviously coincides with the normal to the fiber $\mathbf{n}$, see Fig. 2 a. This implies that $|\nabla H| \rightarrow \infty$ and the normal $\mathbf{N}$ is parallel to the XY -plane. Then, the equality

$$\Gamma_c: \quad \mathbf{J} = \mathbf{N} = \mathbf{n}$$

is satisfied on the contact line Γ_c . As a result, the different forms (14), (25), and (26) of the condition at the boundary Γ gives

$$\Gamma: \quad \theta = \beta, \quad J \equiv J_n = 1, \quad t = 0. \quad (36)$$

At the point D of the boundary CA we have: normal $\mathbf{N}$ is perpendicular to the XY -plane, $J = 0$ and, therefore, $t \rightarrow +\infty$. Taking into account this fact together with the conditions (31), (34), and (36), we conclude that the region Ω_χ in the complex plane of hodograph $\chi = t + i\theta$ is the half-strip, shown in Fig. 3 a.

In order to find the shape of the region Ω_W in the plane of the complex velocity potential $W = -H^{(i)} + i\psi$, we use the boundary conditions (33) and (35). Then, we get the region shown

in Fig. 3 b. It should be note the important feature of all forms (14), (25), (26), and (36) of the condition at the boundary Γ : they contain no information whatever about the behavior of $H^{(i)}$ or ψ . Therefore, Γ is the boundary of unknown shape in the W -plane, and Fig. 3 b shows only the hypothetical shape of this boundary. As a result, we cannot solve the problem of determining the function $W(\chi)$ by using the method of conformal mapping [19].

4.1. Transition to the derivative $dW/d\chi$

We propose to introduce an auxiliary function $F(\chi)$, which is obtained by differentiating $W(\chi)$

$$F(\chi) = \frac{dW}{d\chi}. \quad (37)$$

Let us investigate the boundary behavior of this function. In order to estimate its behavior on the boundary Γ , we introduce the arc length S , which is measured along the contour $\Gamma \in XY$ from the point C , see Fig. 2 b. Taking into account the property (21) of the stream function ψ , we can write the boundary relation on this contour

$$\Gamma : \quad \frac{d\psi}{dS} = J_n. \quad (38)$$

The contour $\Gamma \in XY$ is the unit circle. Then, taking into account formulas (36) and (38), we obtain an important identity for some differentials along this contour

$$\Gamma : \quad d\psi = dS = d\theta = d\beta. \quad (39)$$

Note that the variable t at the boundary Γ in the χ -plane remains constant, see Fig. 3 a. Therefore, we can write the derivative $dW/d\chi$ at this boundary in the form [19]

$$\Gamma : \quad \frac{dW}{d\chi} = \frac{1}{i} \frac{d}{d\theta} \left(-H^{(i)} + i\psi \right) = \frac{d\psi}{d\theta} + i \frac{dH^{(i)}}{d\theta}. \quad (40)$$

As a result, using formulas (39) and (40), we find the behavior of the function $F = dW/d\chi$ at this boundary

$$\Gamma : \quad \operatorname{Re} F = 1. \quad (41)$$

Next, we estimate the behavior of the function $F(\chi)$ on the remaining boundaries AB , DA , and CD . Note that the variable θ at each of these boundaries remains constant, see Fig. 3 a. Similar to (40), we can express the derivative $dW/d\chi$ at these boundaries in the form [19]

$$AB \cup DA \cup CD : \quad \frac{dW}{d\chi} = \frac{d}{dt} \left(-H^{(i)} + i\psi \right) = -\frac{dH^{(i)}}{dt} + i \frac{d\psi}{dt}. \quad (42)$$

As a result, using formulas (33) and (42), we find the behavior of the function $F = dW/d\chi$ at these boundaries

$$AB \cup CD \cup DA : \quad \operatorname{Im} F = 0. \quad (43)$$

We should add to this condition the inequality

$$AB \cup CD \cup DA : \quad \operatorname{Re} F = -\frac{dH^{(i)}}{dt} \geq 0, \quad (44)$$

which is obtained from formula (42) by taking into account the known behavior of the variables $H^{(i)}$ and t at the boundaries $AB \cup DA \cup CD$ in the χ -plane and in the W -plane, see Fig. 3.

Further, by comparing Fig. 3 a with Fig. 3b it is easily established that at the points A and D the conformality property of the mapping $\chi \rightarrow W$ is violated. Then the modulus $|F| = |dW/d\chi|$ at these points is equal to zero or infinity [19]. In the W -plane, we have $|W| \rightarrow \infty$ at point A , and, therefore, in the F -plane we get [19]

$$A : \quad |F| \rightarrow \infty. \quad (45)$$

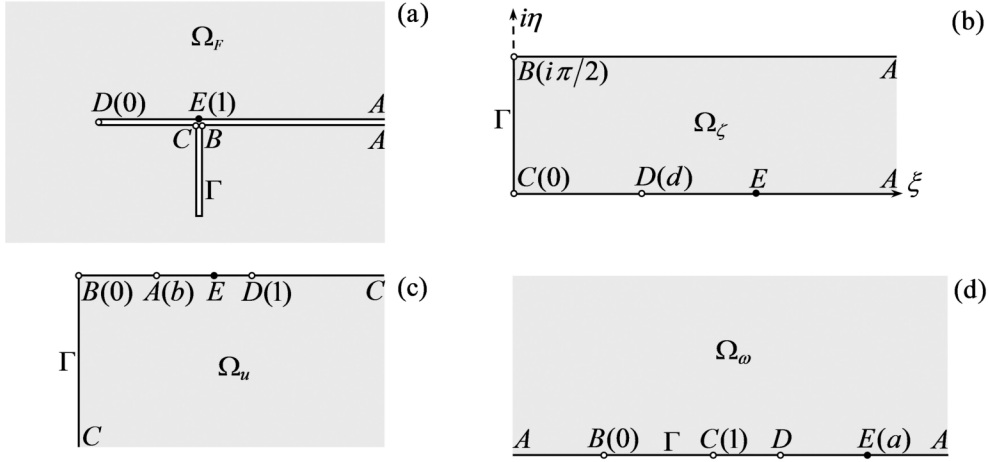


Fig. 4. Complex planes of the auxiliary variables: (a) F ; (b) $\zeta = \xi + i\eta$; (c) u and (d) ω .

Let us turn to the point D . According to Figs. 3 a and b, for the conformal mapping $\chi \rightarrow W$ in the neighborhood of this point, the following estimates hold [19]

$$W \sim W_D : \quad \chi(W) \sim -\frac{1}{2} \ln(W_D - W); \quad \Rightarrow \quad W(\chi) \sim W_D - e^{-2\chi}; \quad \Rightarrow \quad \frac{dW}{d\chi} \sim 2e^{-2t-2i\theta}.$$

At the point D of the χ -plane we have $t \rightarrow +\infty$, see Fig. 3 a. As a result, the function $F = dW/d\chi$ at this point can be evaluated explicitly

$$D : \quad F = 0. \quad (46)$$

Using (41) and (43)–(46), we can estimate the boundary behavior of the function $F(\chi)$. As a result, we conclude that the region Ω_F in the plane of complex variable F is the plane with a polygonal cut, as shown in Fig. 4 a. Regions of this type in the theory of functions of a complex variable are called star-like regions [19]. Note that we additionally introduce the auxiliary point E in Fig. 4a, so that the value $F = 1$ in the region Ω_F now corresponds to three boundary points B , C , and E , each of which lie on different banks of the polygonal cut. We should also note that the length of segment $BC = \Gamma$ in the F -plane is not known in advance. Thus, the regions Ω_χ and Ω_F are defined. It remains to find the function $F(\chi)$ that conformal maps the χ -plane onto the F -plane, so that the function $W(\chi)$ can be obtained by integrating $F(\chi)$ over the variable χ .

5. PARAMETRIZATION AND CONFORMAL MAPPINGS

The shape of the regions Ω_χ and Ω_F is quite simple. Accordingly, we can hope to obtain exact formulas for the conformal mapping $\chi \rightarrow F$. However, it is difficult to write out function $F(\chi)$ directly in an explicit form. Similar to many well-known examples from classical hydrodynamics [20], it is necessary to use some parametrization. By the parametrization, we mean here the introduction of one or more planes of parametric complex variables in order to find the parameterized form of the conformal mappings $\chi \rightarrow F$, using chains of well-known conformal mappings. Here it is convenient to introduce one main plane of the parametric variable ζ and two auxiliary planes of parametric variables u and ω .

We choose the main plane of the parametric variable $\zeta = \xi + i\eta$ in the form of the half-strip Ω_ζ , shown in Fig. 4 b. The differences between the half-strip Ω_ζ and the half-strip Ω_χ are as follows: firstly, the thickness of Ω_ζ is half the thickness of Ω_χ ; and, secondly, infinity in Ω_ζ corresponds to the point A , while in Ω_χ it corresponds to the point D , see Fig. 3 a and Fig. 4 b. Note that the position of the boundary point D in the ζ -plane is specified by an undetermined real parameter $d > 0$.

Further, we choose the auxiliary plane of variable u in the form of the lower right-hand quadrant Ω_u , shown in Fig. 4 c. Note that the position of the boundary point A in the u -plane is specified by an undetermined real parameter b , which lies in the range $0 < b < 1$.

Furthermore, we choose the auxiliary plane of variable ω in the form of the upper half-plane Ω_ω , shown in Fig. 4 d. Note that the position of the boundary point E in the ω -plane is specified by an undetermined real parameter $a > 1$.

Let us explain the motivation for the appropriate choice of the shape of the ζ -, u -, and ω -planes. Using the complex ζ -plane allows us to look for the functions $F(\chi)$ and $W(\chi)$ in a parameterized form, i.e., as a set of three functions $F(\zeta)$, $\chi(\zeta)$ and $W(\zeta)$. First of all, this ζ -plane must be convenient for finding the meniscus shape. Well-known examples of solving similar problems in the case of the vertical thin fiber [5]–[8] shows that our choice of the ζ -plane is convenient for finding the meniscus shape by integrating the parameterized solution on a rectangular grid. In addition, the images of the grid points in the physical plane are concentrated precisely near the fiber, where the meniscus has a complex shape.

The form of the u -plane must provide the known conformal mappings $u \rightleftharpoons \chi$ (direct and inverse). The form of the ω -plane must provide the known conformal mapping $\omega \rightarrow F$. Moreover, all three planes ζ , u , and ω must provide the known conformal mappings $\zeta \rightleftharpoons u$, $\zeta \rightarrow \omega$ and $u \rightleftharpoons \omega$.

Let us write out these known conformal mappings. The quadrant Ω_u (see Fig. 4 c) and the half-strip Ω_χ (see Fig. 3 a) are mapped conformally onto each other by using the functions [13]

$$u(\chi) = \coth \frac{\chi}{2} \quad \Leftrightarrow \quad \chi(u) = \ln \left(\frac{u+1}{u-1} \right). \quad (47)$$

The upper half-plane Ω_ω (see Fig. 4 d) is mapped conformally onto the star-like region Ω_F (see Fig. 4 a) by using the function [21]

$$F(\omega) = 1 + c(\omega - a)\sqrt{\omega(\omega - 1)}, \quad (48)$$

where c is an undetermined real parameter.

The quadrant Ω_u (see Fig. 4 c) and the half-strip Ω_ζ (see Fig. 4 b) are mapped conformally onto each other by using the functions similar to (47)

$$u(\zeta) = b \coth \zeta \quad \Leftrightarrow \quad \zeta(u) = \frac{1}{2} \ln \left(\frac{u+b}{u-b} \right). \quad (49)$$

The same half-strip Ω_ζ (see Fig. 4 b) is mapped conformally onto the upper half-plane Ω_ω (see Fig. 4 d) by using the function [13]

$$\omega(\zeta) = \cosh^2 \zeta. \quad (50)$$

The quadrant Ω_u (see Fig. 4 c) and the upper half-plane Ω_ω (see Fig. 4 d) are mapped conformally onto each other by using the functions [19]

$$u(\omega) = b \sqrt{\frac{\omega}{\omega - 1}} \quad \Leftrightarrow \quad \omega(u) = \frac{u^2}{u^2 - b^2}. \quad (51)$$

Now, using the second expression of (47) and the first expression of (49), we can find the exact expression for $\chi(\zeta)$

$$\chi(\zeta) = \ln \left(\frac{b \coth \zeta + 1}{b \coth \zeta - 1} \right). \quad (52)$$

Using formulas (48) and (50), we can find the exact expression for $F(\zeta)$:

$$F(\zeta) = 1 + c \sinh \zeta \cosh \zeta (\cosh^2 \zeta - a). \quad (53)$$

Knowing functions $\chi(\zeta)$ and $F(\zeta)$, we can also find the function $W(\zeta)$. Indeed, taking into account (37), we can write the relation

$$F(\zeta) = \frac{dW}{d\zeta} \left(\frac{d\chi}{d\zeta} \right)^{-1}. \quad (54)$$

The derivative $d\chi/d\zeta$ can be found by direct differentiation of (52):

$$\frac{d\chi}{d\zeta} = \frac{2b}{b^2 \cosh^2 \zeta - \sinh^2 \zeta} = \frac{2b}{1 - (1 - b^2) \cosh^2 \zeta}. \quad (55)$$

If we substitute the formulas (53) and (55) into (54), we obtain

$$\frac{dW}{d\zeta} = \frac{d\chi}{d\zeta} + 2bc \frac{\sinh \zeta \cosh \zeta (\cosh^2 \zeta - a)}{1 - (1 - b^2) \cosh^2 \zeta}. \quad (56)$$

The right-hand side of this expression can be integrated [22]. As a result, we find the explicit form of the function $W(\zeta)$

$$W(\zeta) = \chi(\zeta) - \frac{bc}{(1 - b^2)} \left\{ \cosh^2 \zeta + \left[\frac{1}{(1 - b^2)} - a \right] \ln [1 - (1 - b^2) \cosh^2 \zeta] \right\} - (H_B + i\pi). \quad (57)$$

Here H_B is an integration constant, which is equal to the height of meniscus $Z = H$ at the point B , where $\chi = i\pi$ and $\cosh^2 \zeta = \omega = 0$.

Note that the formulas (47)–(49), (51), (52), and (57) involve the multivalued functions, namely the logarithm and the square root [19]. Using these functions, we choose their single-valued branches that are real and positive on the boundary CD .

Thus, we have found the exact analytical form of two functions $\chi(\zeta)$ and $W(\zeta)$, which together represent the desired solution to the problem of the internal asymptotic expansion (13)–(16). But in the course of its construction we had to introduce several auxiliary parameters so far undetermined.

6. DETERMINATION OF AUXILIARY PARAMETERS

The direct problem of the inner asymptotic expansion (13)–(16) depends only on the physical parameter α_0 (the angle of fiber inclination), since we fixed the second physical parameter, namely the contact angle: $\gamma_0 = 0$. However, we have found not a direct but some parameterized solution of the problem. This solution depends on several auxiliary parameters, namely a , b , c , d , and H_B , which are now to be determined. There is also parameter Q , but we already know its relationship with α_0 , see (32).

In our parameterized solution, we propose to choose the parameter b as the main auxiliary parameter of the problem, since its relationship with the physical parameter α_0 can be easily established. Indeed, at the point A we have $u = b$. Taking into account the first expression of (47), we can write the identity

$$A: \quad b = u = \coth \left(\frac{t_A + i\pi}{2} \right) = \frac{\sinh t_A}{1 + \cosh t_A}.$$

Then, using the second expression of (31), we can establish such a relationship between b and α_0

$$b = \frac{\cos \alpha_0}{1 + \sin \alpha_0} = \frac{1 - \sin \alpha_0}{\cos \alpha_0} \quad \Leftrightarrow \quad \sin \alpha_0 = \frac{1 - b^2}{1 + b^2}, \quad \cos \alpha_0 = \frac{2b}{1 + b^2}. \quad (58)$$

Note that in the limit $\alpha_0 \rightarrow 0$ we have $b \rightarrow 1$.

All other auxiliary parameters of the problem, namely a , c , d , and H_B , should be expressed through the main auxiliary parameter b .

6.1. Expression of the parameters a , c and d as functions of b

The expression for the parameter d can be found by comparing the position of the point D in the ζ - and u -planes, see Figs. 4 b and c. Using the second relation of (49), we obtain

$$d(b) = \frac{1}{2} \ln \left(\frac{1 + b}{1 - b} \right). \quad (59)$$

Some relationships between the parameters can be established due to the properties of the conformal mapping $\omega \rightarrow F$ of the star-like region Ω_F onto the upper half-plane Ω_ω [21]. Let ω_D

denote the position of the point D in the ω -plane. The expression ω_D as a function of b can be obtained by comparing the position of the point D in the u - and ω -planes, see Fig. 4c and Fig. 4d. Using the second relation of (51), we obtain

$$\omega_D = (1 - b^2)^{-1}. \quad (60)$$

At the same time, the parameter ω_D must satisfy the equation [21]

$$\frac{1}{[F(\omega) - 1]} \frac{dF}{d\omega} \Big|_{\omega=\omega_D} = 0.$$

Using the form (48) of the function $F(\omega)$, we arrive at the equation

$$4\omega_D^2 - 3\omega_D - 2a\omega_D + a = 0.$$

We can solve this equation for a . Then, using (60), we can obtain the expression for the parameter a :

$$a(b) = \frac{4\omega_D^2 - 3\omega_D}{2\omega_D - 1} = \frac{1 + 3b^2}{1 - b^4}. \quad (61)$$

The expression for the parameter c can be found by comparing the position of the point D in the F - and ω -planes, see Figs. 4 a and d. Using the form (48) of the function $F(\omega)$, we obtain the equation

$$1 + c(\omega_D - a)\sqrt{\omega_D(\omega_D - 1)} = 0.$$

We can solve this equation for c . Then, using (60) and (61), we obtain the expression $c(b)$:

$$c(b) = \frac{(1 - b^4)(1 - b^2)}{2b^3}. \quad (62)$$

Thus, we have found the expressions for the parameters a , c and d as functions of b . Note that the form of the function $W(\zeta)$, formula (57), involves the auxiliary parameters a , b , and c . Now we can get rid of the parameters a and c . Indeed, from (61) and (62) we have the following two identities

$$\frac{bc}{(1 - b^2)} = \frac{1 - b^4}{2b^2}, \quad \frac{1}{(1 - b^2)} - a = -\frac{2b^2}{1 - b^4}.$$

Using these identities, one can significantly simplify the right-hand expression of (57)

$$W(\zeta) = \chi(\zeta) - \frac{(1 - b^4)}{2b^2} \cosh^2 \zeta + \ln [1 - (1 - b^2) \cosh^2 \zeta] - (H_B + i\pi), \quad (63)$$

where, instead of the multivalued logarithm function, one should choose such a single-valued branch that is real and positive at the boundary CD . Here the integration constant H_B also appears. Its definition constitutes the second part of this section.

6.2. Determination of the integration constant H_B

This constant is involved in the function $H^{(i)}(\xi, \eta) = -\text{Re } W(\zeta)$. It seems that the most natural approach to determine the constant H_B is this: we find the value of the function $H^{(i)}(\xi, \eta)$ at any of the boundary points A , B , C , and D ; then we compare this value with the known value of the meniscus height $H(X, Y)$ at the same point, provided that the point lies in the zone $R \sim 1$, where the inner asymptotic expansion is valid. But in this zone there is not a single point of the boundary with a known meniscus height $H(X, Y)$! The same can be said about the meniscus height $h(x, y)$ in the xyz -coordinate system. The only value of the function $h(x, y)$ that we know is the value $h = 0$ at infinity $r \rightarrow \infty$. But at infinity the inner asymptotic expansion is not valid!

Therefore, we will have to determine the constant H_B , using a rather complicated approach. Firstly, for this purpose we can use only the condition at infinity (6) of the original boundary problem (3)–(6). Secondly, since we have made the transition to the new XYZ -coordinate system

and to the new function $H(X, Y)$, the condition (6) for the function $h(x, y)$ should be rewritten as some condition for the function $H(X, Y)$ at infinity $R \gg 1$. And thirdly, this rewritten condition (6) can be applied to the function $H^{(i)}(X, Y)$ only indirectly, by linking $H^{(i)}(X, Y)$ with the function $H^{(u)}(X, Y)$, which uniformly approximates the meniscus Σ everywhere.

Following this scheme, from (6) we obtain that the function $h^{(u)}(x, y)$ must satisfy the condition $h^{(u)} \rightarrow 0$ at infinity $r \rightarrow \infty$. In terms of the function $H^{(u)}(X, Y)$, this leads to the condition

$$R \gg 1 : \quad H^{(u)}(X, Y) = X \tan \alpha_0 + o(1), \quad (64)$$

according to formula (10). Note that on the right-hand side of (64) the term of order 1, i.e., $O(R^0)$, is equal to zero.

Next, we already have the expression (18), which links the functions $H^{(i)}(X, Y)$ and $H^{(u)}(X, Y)$. It should be noted that the modified Kelvin function K_0 involved in this expression disappears at infinity [14]. Then, we can make sure that condition (64) for the function $H^{(u)}(X, Y)$ turns into the condition (16) for the function $H^{(i)}(X, Y)$. As a result, the following conclusion can be made: in order to determine the constant of integration H_B , we can use the condition at infinity (16) taking into account all terms of order 1, i.e., $O(R^0)$. A similar conclusion was made in the case of the vertical fiber [5].

Before we apply the condition (16), it can be simplified as follows: instead of the neighborhood of infinity in its general form $R \gg 1$, we restrict ourselves to its particular form $Y = 0, X \gg 1$. Using this simplification as well as the formula (32), we transform the condition (16) into the following condition

$$Y = 0, X \gg 1 : \quad H^{(i)}(X) = X \tan \alpha_0 - \frac{1}{\cos^2 \alpha_0} \left[\ln \left(\frac{X}{\cos \alpha_0} \right) + C_\varepsilon \right] + O(X^{-1} \ln X). \quad (65)$$

Thus, we have the estimate for the behavior of the boundary function $H^{(i)}(X)$ on DA in the neighborhood of the point A in the XY -plane.

On the other hand, we have the exact expression (63) for the function $W(\zeta)$, and therefore we can obtain the same exact expressions for the functions $W(\chi)$, $W(u)$, etc. All these functions will contain the desired constant H_B and consequently any of them can be used in our analysis. It seems most convenient to use the function $W(u)$. To find its exact form, we take into account that from the combination of the relation (50) and the second relation in (51) follows the relation

$$\cosh^2 \zeta = \frac{u^2}{u^2 - b^2}.$$

Substituting this relation as well as the second relation from (47) into the expression (63), we obtain such an exact expression for $W(u)$ in the whole u -plane

$$W(u) = \ln \frac{b^2(u+1)^2}{u^2 - b^2} - \frac{1 - b^4}{2(u^2 - b^2)} - \left(H_B + \frac{1 - b^4}{2b^2} + i\pi \right). \quad (66)$$

Taking the real part of (66) and setting $u \in CDA$, we obtain an exact estimate of the behavior of the boundary function $H^{(i)}(u) = -\operatorname{Re} W(u)$ on the entire boundary CDA including the neighborhood of the point A in the u -plane

$$CDA : \quad H^{(i)}(u) = \ln \frac{u^2 - b^2}{(u+1)^2} + \frac{1 - b^4}{2(u^2 - b^2)} + \left(H_B + \frac{1 - b^4}{2b^2} - 2 \ln b \right), \quad \operatorname{Im} u = 0. \quad (67)$$

The desired constant H_B can be determined by equating the estimates (65) and (67), obtained for the boundary functions $H^{(i)}(X)$ and $H^{(i)}(u)$ on DA in the neighborhood of point A . But for this, we need to additionally use one of the two boundary functions at least asymptotically in the neighborhood of the point A : either $u(X)$ on DA in the XY -plane; or $X(u)$ on DA in the u -plane. Accordingly, we can propose *two ways* for finding the constant H_B .

The *first way* is based on the use of the boundary function $u(X)$ on DA in the XY -plane. This function can be estimated using the variable J , i.e., the magnitude of the fictitious flow. Variables J and t are related by the one-to-one correspondence (22). Since $t = \operatorname{Re} \chi$ and the mappings $u \rightleftharpoons \chi$ have already been found, see formula (47), we can easily establish the exact expression for the boundary function $u(J)$ on DA .

On the other hand, by differentiating the boundary relation (65) with respect to X , we can find $\partial H^{(i)}/\partial X$ in the neighborhood of infinity $Y = 0$, $X \gg 1$. Accordingly, one can asymptotically estimate $\nabla H^{(i)}$ on DA in the neighborhood of the point A in the XY -plane, since we have $\partial H^{(i)}/\partial Y = 0$ on DA , see (15). If we know $|\nabla H^{(i)}|$, then we can uniquely determine the magnitude of the fictitious flow J , see (20). Therefore, one can asymptotically estimate the boundary behavior of the function $J(X)$ on DA in the neighborhood of the point A . Finally, by substituting this estimate into the exact expression for $u(J)$, we can obtain the desired estimate for the behavior of the boundary function $u(X)$ on DA in the neighborhood of the point A in the XY -plane. So, for example, the integration constant was found in [6].

However, in our case, the behavior of the boundary function $H^{(i)}(X)$ on DA at infinity, according to the estimate (65), is determined not by the singularity $\sim \ln X$, as in the case of the vertical fiber [5], [6], but by the stronger singularity $\sim X$. Therefore, the terms of the asymptotic expansion (65) of the function $H^{(i)}(X)$ for small X^{-1} are not sufficient: it is necessary to find higher-order terms of the expansion (65). In our case, this seems to be quite a difficult problem.

The *second way* for finding the constant H_B is based on the use of the boundary function $X(u)$ on DA in the u -plane. This function can be estimated using direct integration of Khristianovich's differential relations (24). In general, asymptotic analysis of such integral for small X^{-1} can also be a difficult problem, but in our case this integral can be calculated exactly. As a result, it is possible to obtain the exact form of the boundary function $X(u)$ on the entire boundary CDA . The details are given in the *Appendix*, here we present only the final form of this function

$$CDA: \quad X(u) = 1 - \frac{1-b^2}{2b} \ln \frac{u-b}{u+b} - \frac{2}{u+1} + \frac{u(1+b^2)}{u^2-b^2}. \quad (68)$$

Substituting this formula into the expression (65), we obtain the first estimate of the boundary function $H^{(i)}(u)$ on DA in the neighborhood of the point A in the u -plane. The second estimate of this function is given directly by the expression (67), which involve H_B . Comparing these two estimates, we obtain (see *Appendix*)

$$H_B = -\frac{(1-b)(1+3b-b^2+b^3)}{4b^2} + \frac{1+b^4}{4b^2} \ln \frac{8b^2}{(1+b^2)^2} + \ln \frac{b(1+b)^2}{\sqrt{2}(1+b^2)} - \frac{(1+b^2)^2}{4b^2} C_\varepsilon, \quad (69)$$

where the constant C_ε is determined by the formula (17).

7. CONFIGURATION OF EXTERNAL MENISCI

First, we find the surface $Z = H^{(i)}(X, Y)$, which corresponds to the obtained solution for the problem of the inner asymptotic expansion (i.e., which approximates the desire meniscus Σ only in zone $R \sim 1$). This solution was parameterized by introducing the auxiliary complex plane of parametric variable $\zeta = \xi + i\eta$. Accordingly, it is reasonable to look for the surface $Z = H^{(i)}(X, Y)$ as some parametric surface, i.e., we need to find three functions $H^{(i)}(\xi, \eta)$, $X(\xi, \eta)$ and $Y(\xi, \eta)$ [11].

The first of these functions, namely $H^{(i)}(\xi, \eta)$, can be found by the formula

$$H^{(i)}(\xi, \eta) = -\operatorname{Re} W(\zeta), \quad (70)$$

where the function $W(\zeta)$ is expressed by the exact formulas (63) and (69). The remaining functions $X(\xi, \eta)$ and $Y(\xi, \eta)$ can be determined using the differential relations

$$\frac{\partial}{\partial \xi}(X+iY) = e^{i\theta} \left[i \cosh t \frac{\partial \psi}{\partial \xi} - \sinh t \frac{\partial H^{(i)}}{\partial \xi} \right], \quad \frac{\partial}{\partial \theta}(X+iY) = e^{i\theta} \left[i \cosh t \frac{\partial \psi}{\partial \eta} - \sinh t \frac{\partial H^{(i)}}{\partial \eta} \right], \quad (71)$$

which follow from Khristianovich's formulas (24). Here the complex variable $X + iY$ is used only to shorten the expressions (i.e., we do not imply any functions of the complex variable $X + iY$).

The partial derivatives appearing in (71) are related to the derivative $dW/d\zeta$ as follows [19]

$$\frac{dW}{d\zeta} = -\frac{\partial H^{(i)}}{\partial \xi} + i\frac{\partial \psi}{\partial \xi} = \frac{\partial \psi}{\partial \eta} + i\frac{\partial H^{(i)}}{\partial \eta}.$$

Then, taking into account the formula (71), we can obtain the functions $X(\xi, \eta)$ and $Y(\xi, \eta)$ by using line integrals of vector fields

$$\begin{aligned} X(\xi, \eta) + iY(\xi, \eta) = 1 + \int_0^\zeta e^{i\theta} \sinh t \left[\operatorname{Re} \left\{ \frac{dW}{d\zeta} \right\} d\xi - \operatorname{Im} \left\{ \frac{dW}{d\zeta} \right\} d\eta \right] \\ + i \int_0^\zeta e^{i\theta} \cosh t \left[\operatorname{Im} \left\{ \frac{dW}{d\zeta} \right\} d\xi + \operatorname{Re} \left\{ \frac{dW}{d\zeta} \right\} d\eta \right], \end{aligned} \quad (72)$$

where the formulas (52) and (56) should be taken into account. In deriving (72), we have used that $X_C = 1$ and $Y_C = 0$.

If we use a rectangular grid $\{\xi_k, \eta_m\}$ in the ζ -plane, then in formula (72) we can pass from line integrals to ordinary ones, i.e., to integrals either over ξ or over η . Indeed, let us introduce into the ζ -plane a rectangular grid with a uniform step of size $\Delta\xi > 0$ along ξ -axis and with a uniform step of size $\Delta\eta > 0$ along η -axis:

$$\xi_k = (k-1)\Delta\xi, \quad k = 1, \dots, K; \quad \eta_m = \frac{\pi(m-1)}{2(M-1)}, \quad m = 1, \dots, M.$$

To ensure good quality of drawing the meniscus in the vicinity of point D , the step size $\Delta\xi$ should be chosen so that the value of parameter d , corresponding to this point in the ζ -plane, coincides with one of the grid point ξ'_k , see formula (59) and Fig. 4 b

$$D: \quad \eta = 0, \quad \xi'_k = d.$$

Using our rectangular grid, the three functions $H^{(i)}(\xi, \eta)$, $X(\xi, \eta)$, and $Y(\xi, \eta)$ can be matched to three two-dimensional arrays $H_{km}^{(i)}$, X_{km} and Y_{km} . The two-dimensional array $H_{km}^{(i)}$ corresponding to the function $H^{(i)}(\xi, \eta)$ is determined by the formula (70)

$$H_{km}^{(i)} = -\operatorname{Re} W(\zeta)|_{\zeta=\xi_k+i\eta_m}, \quad k = 1, \dots, K; \quad m = 1, \dots, M, \quad (73)$$

where the formulas (63) and (69) should be used.

The two-dimensional arrays X_{km} and Y_{km} corresponding to functions $X(\xi, \eta)$ and $Y(\xi, \eta)$ can be found as follows. First, by choosing the contour Γ : $\xi = 0$, $\eta \in [0, \pi/2]$ in the ζ -plane as the integration path in the formula (72), we can find the shape of this contour in the XY -plane

$$\Gamma = BC: \quad X_{1m} + iY_{1m} = 1 + \int_0^{\eta_m} e^{i\theta} \left[i \cosh t \operatorname{Re} \left\{ \frac{dW}{d\zeta} \right\} - \sinh t \operatorname{Im} \left\{ \frac{dW}{d\zeta} \right\} \right] \Big|_{\xi=0} d\eta, \quad (74)$$

where $m = 1, \dots, M$. We make sure that the found set of points (X_{1m}, Y_{1m}) actually forms the unit semicircle $\Gamma = BC$ in the XY -plane, see Fig. 2 b. Then, we can find the remaining members of the arrays X_{km} and Y_{km} by choosing the straight lines $\eta = \eta_m$ in the ζ -plane with the origin at the points of the boundary $\Gamma = BC$ as the integration path in the formula (72)

$$X_{km} + iY_{km} = X_{1m} + iY_{1m} + \int_0^{\xi_k} e^{i\theta} \left[\sinh t \operatorname{Re} \left\{ \frac{dW}{d\zeta} \right\} + i \cosh t \operatorname{Im} \left\{ \frac{dW}{d\zeta} \right\} \right] \Big|_{\eta=\eta_m} d\xi, \quad (75)$$

where $k = 2, \dots, K$ and $m = 2, \dots, M$.

The boundary CDA , which corresponds both to the straight line $\eta = 0$ in the ζ -plane and the index $m = 1$ of the rectangular grid, must be examined separately. On this boundary we have: the

angle θ is equal to 0 or π by virtue of (34), and the derivative $dW/d\zeta$ is real, see Fig. 3 b and Fig. 4 b. If we use the formula (75), we get

$$CDA: \quad X_{k1} = 1 + \int_0^{\xi_k} \cos \theta \sinh t \frac{dW}{d\zeta} \Big|_{\eta=0} d\xi, \quad Y_{k1} = 0; \quad k = 1, \dots, K. \quad (76)$$

The path of integration in the integral (76) passes through the point D : $\xi = d = \xi'_k$, at which $J = 0$ and, therefore, $t \rightarrow \infty$, by virtue of (22). Thus, the hyperbolic sine involved in the integrand has a singularity at the point D : $\sinh t \rightarrow \infty$. Estimates show that the remaining multiplicative part of the integrand at the point D must be equal to zero, and the integrand itself at this point must be integrable. However, applying standard formulas for numerical integration to the integral (76) leads to large errors in the vicinity of the point D . These difficulties can be avoided by using the exact expression (68) for the boundary function $X(u)$ on CDA . Then, taking into account the first expression in (49), instead of the formula (76), we obtain the formula

$$CDA: \quad X_{k1} = X(u)|_{u=b \coth \xi_k}, \quad Y_{k1} = 0; \quad k = 1, \dots, K, \quad (77)$$

where for $X(u)$ the exact expression (68) should be used.

We have obtained the parametric equation (73)–(75), (77) for the surface $Z = H^{(i)}(X, Y)$, which approximates the desired meniscus Σ in the zone of the inner asymptotic expansion $R \sim 1$. Now, using the formulas (18) and (32), we can find the members of the array $H_{km}^{(u)}$

$$H_{km}^{(u)} = H_{km}^{(i)} + \frac{1}{\cos^2 \alpha_0} \left[K_0 \left(\varepsilon^{1/2} \sqrt{\frac{X_{km}^2}{\cos^2 \alpha_0} + Y_{km}^2} \right) + \frac{1}{2} \ln \left(\frac{X_{km}^2}{\cos^2 \alpha_0} + Y_{km}^2 \right) + C_\varepsilon \right], \quad (78)$$

which corresponds to the function $H^{(u)}(X, Y)$. Expression (78) together with the formulas (73)–(75), (77) represents the parametric equation of surface $Z = H^{(u)}(X, Y)$, which uniformly approximates the desired meniscus Σ everywhere in the XY -plane.

Finally, using the formulas (7) and (8) of the coordinate transformation $XYZ \rightarrow xyz$, we can return to the original Cartesian coordinate system xyz :

$$y_{km} = Y_{km}; \quad x_{km} = X_{km} \cos \alpha_0 + H_{km}^{(u)} \sin \alpha_0; \quad h_{km}^{(u)} = -X_{km} \sin \alpha_0 + H_{km}^{(u)} \cos \alpha_0, \quad (79)$$

where $k = 1, \dots, K$ and $m = 1, \dots, M$.

The expression (79) together with the formulas (73)–(75), (77) and (78) represents the parametric equation of the surface $z = h^{(u)}(x, y)$, which uniformly approximates the desired meniscus Σ everywhere in the xy -plane.

8. ANALYSIS OF RESULTS

Since we are dealing with an asymptotic formulation of the problem for small Bond number, to illustrate specific example it is necessary to choose a specific value of this number. We choose $\varepsilon = 0.01$ so that the error of the uniform asymptotics (79) would be of the order 0.01.

Figure 5 shows examples of the configuration of the external meniscus Σ for different values of the inclination angle α_0 of the circular fiber: a) $\alpha_0 = 0$, b) $\alpha_0 = \pi/6$, c) $\alpha_0 = \pi/4$, and d) $\alpha_0 = \pi/3$. The level curves of constant height h are shown in Fig. 5 as the white lines with an interval of $(h_{\max} - h_{\min})/23$, where $h_{\min} = \min[h^{(u)}(x, y)]$ and $h_{\max} = \max[h^{(u)}(x, y)]$. The contact line Γ_c is shown as the black line. In the case of the vertical fiber ($\alpha_0 = 0$), the external meniscus Σ is characterized by axial symmetry. At the same time, the inclination of the fiber from the vertical leads to an asymmetry of Σ , which increases with increasing angle of fiber inclination from the vertical.

As seen in Fig. 5, when the fiber is tilted to the right, it is the right side of the contact line Γ_c that rises higher than in the case of the vertical fiber. At the same time, the opposite left side of Γ_c falls lower than in the case of the vertical fiber. It should be noted that such a pattern of meniscus asymmetry differs from our intuitive expectations, see Fig. 1 a.

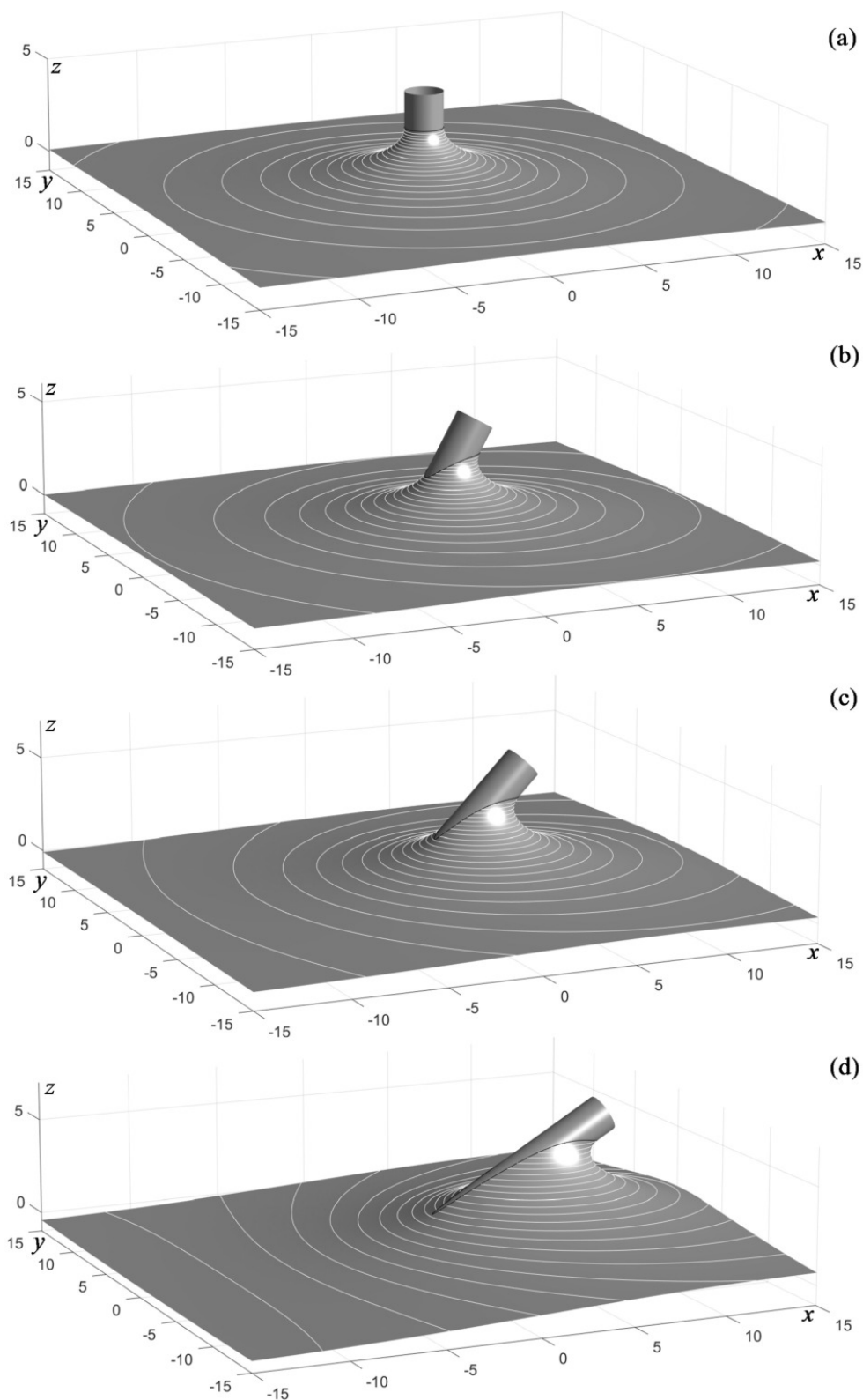


Fig. 5. Examples of the configuration of the external meniscus Σ for $\varepsilon = 0.01$ and for different values of the inclination angle α_0 of the circular fiber: a) $\alpha_0 = 0$, b) $\alpha_0 = \pi/6$, c) $\alpha_0 = \pi/4$, d) $\alpha_0 = \pi/3$.

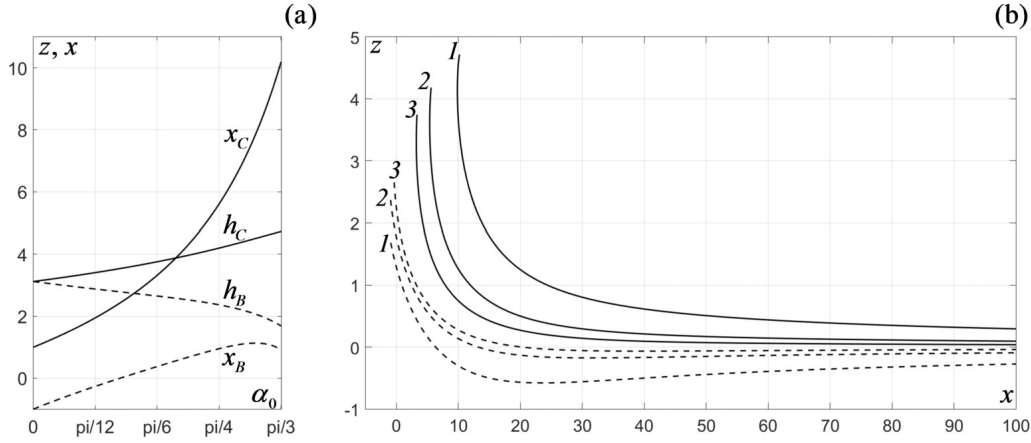


Fig. 6. (a) Heights h and abscissas x of the points C and B depending on the value of α_0 . (b) Graphs $z = h_{CA}(x)$ (solid lines) and $z = h_{BA}(-x)$ (dashed lines) correspond to the curves CA and BA for $\varepsilon = 0.01$ and for three values of α_0 : 1 for $\alpha_0 = \pi/3$, 2 for $\alpha_0 = \pi/4$, 3 for $\alpha_0 = \pi/6$.

The meniscus asymmetry can be quantitatively assessed by comparing both the heights and the abscissas of the points C and B depending on the value of α_0 , see Fig. 6 a, where the solid lines are the graphs $h_C(\alpha_0)$ and $x_C(\alpha_0)$; the dashed lines are the graphs $h_B(\alpha_0)$ and $x_B(\alpha_0)$.

Note that the lowering of the left side of the contact line Γ_c leads to the lowering of some part of the meniscus Σ , and this part falls even below than the level $z = 0$ of the initially undisturbed liquid. This is easy to see in Fig. 5 d for $\alpha_0 = \pi/3$, but we maintain that, within the framework of our asymptotic formulation of the problem, the meniscus always has this qualitative feature for any $\alpha_0 > 0$. For large α_0 the part of the meniscus, which falls below the level $z = 0$, is located quite close to the fiber; the magnitude of the smallest lowering of the meniscus $|h_{min}|$ is quite large; and, hence, the lowering itself is easy to see. With decreasing α_0 , the value of $|h_{min}|$ decreases rapidly; the part of the meniscus, which falls below the level $z = 0$, moves further and further away from the fiber; and, hence, it becomes more difficult to see the lowering itself.

To illustrate this effect, in Fig. 6 b we show the curves CA (solid lines) and BA (dashed lines) as the cross-sections of the meniscus Σ by the vertical xz -plane for three values of α_0 : 1 – $\alpha_0 = \pi/3$, 2 – $\alpha_0 = \pi/4$, 3 – $\alpha_0 = \pi/6$. It is clear that all of these curves tend to zero at infinity, but if on the right side the curve CA approaches zero from above, then on the left side the curve BA approaches zero from below!

One more interesting result can be noted: although the highest point of the contact line Γ_c lies near the point C , it does not coincide with this point itself. This is clearly demonstrated in Fig. 7, where for $\alpha_0 = \pi/4$ we present projections of the contact line Γ_c : a) onto the xz -plane; b) onto the yz -plane; and c) onto the xy -plane.

Note that we limited our analysis of examples to the maximum value of the inclination angle of the fiber $\alpha_0 = \pi/3$. The exact formulas of the obtained solution allow us to find meniscus configurations for the inclination angles greater than $\pi/3$, up to $\pi/2$. All noted trends will manifest themselves even strongly. However, at angles $\alpha_0 > \pi/3$, some doubts arise about the validity of the asymptotic estimates. Indeed, we estimate the asymptotic parameter ε based on the characteristic linear scale L_m , which we have chosen to be equal to half the fiber diameter. But it is more correct to associate the characteristic linear scale not with the fiber diameter, but with the size of the projection of the contact line Γ_c onto the horizontal xy -plane. It is obvious that the maximum size of Γ_c in the y -direction is always exactly 2 (fiber diameter). At the same time, the maximum size of Γ_c in the x -direction, i.e., value $(x_C - x_B)$, is already approximately 9.3 for $\alpha_0 = \pi/3$, see Fig. 6a. This is approximately 4.6 times the fiber diameter, and we believe that such an increase of the real characteristic scale is already critical. Therefore, the inclination angles $\alpha_0 > \pi/3$ are not interesting.

Note that the effect of local lowering of the meniscus below level $z = 0$ of the initially undisturbed liquid for any inclination angle is quite unexpected. It would be interesting to confirm or disprove this, say, numerically or experimentally.

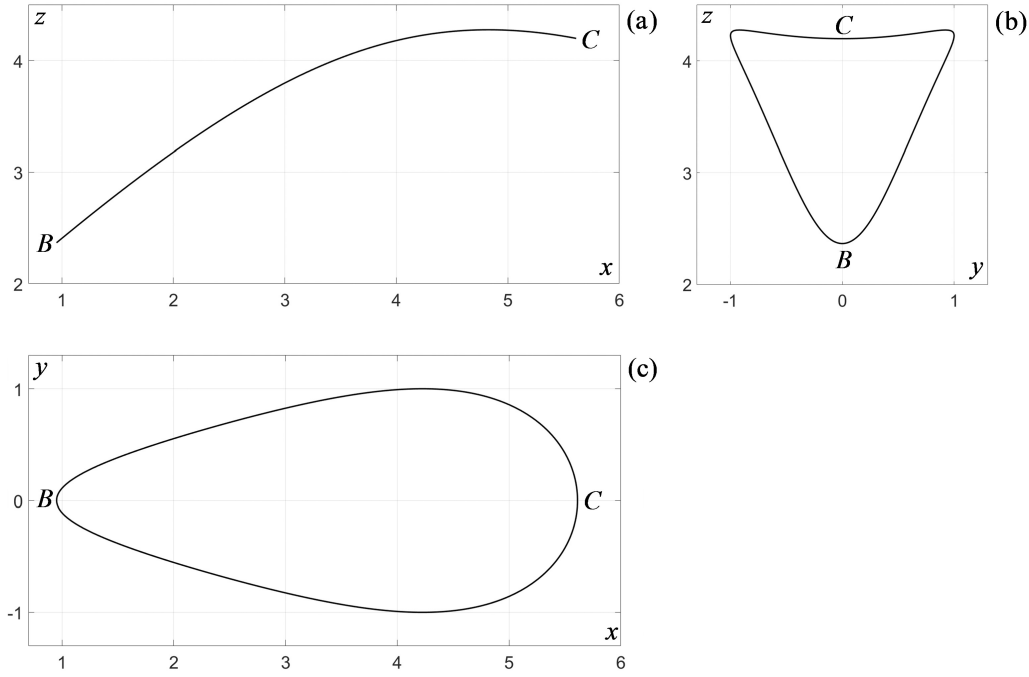


Fig. 7. Projections of the contact line Γ_c for $\varepsilon = 0.01$ and for $\alpha_0 = \pi/4$: (a) onto the xz -plane; (b) onto the yz -plane, and (c) onto the xy -plane.

CONCLUSIONS

Using a new asymptotic formulation, we construct an analytical solution for the problem of determining the configuration of the external meniscus on a thin circular fiber, which is immersed in a liquid at an arbitrary inclination angle to the vertical, under the assumption that the liquid completely wets the fiber. It is shown that the contact line becomes asymmetrical: on the side to which the fiber is inclined, the contact line rises higher, and on the other side it falls lower than in the case of the vertical fiber. This asymmetry increases with increasing angle of the fiber inclination to the vertical. Although wetting causes capillary rise of the liquid on the fiber, it is found that for the inclined fiber some part of the meniscus falls even below the horizontal level of the initially undisturbed liquid. Within the framework of the accepted asymptotic formulation of the problem, this effect occurs at any inclination angle different from zero.

APPENDIX

1. Let us find the exact form of the boundary function $X(u)$ on CDA using Khristianovich's differential relations (24). Note that the boundary CDA is a streamline, and that the angle θ on CDA is equal to 0 or π , see the conditions (28) and (29). Then, writing out the first relation of (24) on this boundary, we obtain

$$CDA: \quad dX = -\cos \theta \sinh t \, dH^{(i)} = \sinh \chi \, dW.$$

The desired boundary function $X(u)$ on CDA can be obtained by integrating this boundary relation along the boundary itself

$$CDA: \quad X(u) = 1 + \int_{\infty}^u \sinh \chi(u) \frac{dW}{du} du, \quad \text{where } \text{Im } u = 0. \quad (80)$$

Here it is taken into account that $X_C = 1$ and that the variable u on CDA is real, see Fig. 4c.

Let's look at the integrand in (80). The function $\chi(u)$ has already been found, see the second expression in (47). Accordingly, we can also find an expression for $\sinh \chi(u)$

$$\sinh \chi(u) = \frac{1}{2} [e^{\chi(u)} - e^{-\chi(u)}] = \frac{1}{2} \left[\frac{u+1}{u-1} - \frac{u-1}{u+1} \right] = \frac{2u}{(u^2-1)} \quad (81)$$

The derivative dW/du can be found by differentiating the expression (66)

$$\frac{dW}{du} = \frac{2}{(u+1)} - \frac{2u}{u^2-b^2} + \frac{u(1-b^4)}{(u^2-b^2)^2}. \quad (82)$$

We substitute now the expressions (81) and (82) into the formula (80). As a result, we get

$$CDA: \quad X(u) = 1 - 2I(u)|_{\infty}^u, \quad \text{where } \operatorname{Im} u = 0. \quad (83)$$

Here $I(u)$ denotes the integral of a rational function

$$I(u) = - \int \frac{u}{(u^2-1)} \left[\frac{2}{(u+1)} - \frac{2u}{u^2-b^2} + \frac{u(1-b^4)}{(u^2-b^2)^2} \right] du = \int u \frac{2u^2 + (1+b^2)^2 u + 2b^4}{(u+1)^2 (u^2-b^2)^2} du. \quad (84)$$

This integral can be calculated exactly in closed form [22]. Omitting the details, we present the final result

$$I(u) = \frac{1-b^2}{4b} \ln \left(\frac{u-b}{u+b} \right) + \frac{1}{u+1} - \left(\frac{1+b^2}{2} \right) \frac{u}{u^2-b^2}.$$

The validity of this expression can be easily verified by directly differentiating it and by comparing the result with the formula (84).

Substituting the found expression for $I(u)$ into the formula (83), we find the exact expression for the desired boundary function $X(u)$ on CDA

$$CDA: \quad X(u) = 1 - \frac{1-b^2}{2b} \ln \left(\frac{u-b}{u+b} \right) - \frac{2}{u+1} + \frac{u(1+b^2)}{u^2-b^2}. \quad (85)$$

Using this exact expression, in particular, one can obtain an estimate of the boundary function $X(u)$ on DA in the neighborhood of point A in the u -plane. The point A itself in the u -plane corresponds to the value $u=b$, and the neighborhood of this point on the boundary DA corresponds to the interval $0 < u-b < \delta$, where $\delta \ll 1$. Then, from (85) follows the estimate

$$0 < u-b < \delta: \quad X(u) = \frac{1+b^2}{2(u-b)} - \frac{1-b^2}{2b} \ln(u-b) + C_X + O(\delta), \quad (86)$$

where C_X denotes such a constant

$$C_X = \frac{1+b^2}{4b} - \frac{1-b}{1+b} + \frac{1-b^2}{2b} \ln(2b). \quad (87)$$

2. Let us substitute the found estimate (86) into the expression (65) for the function $H^{(i)}(X)$. We obtain the following estimate of the boundary function $H^{(i)}(u)$ on DA in the neighborhood of point A in the u -plane

$$H^{(i)}(u) = \frac{(1+b^2) \tan \alpha_0}{2(u-b)} + \left[\frac{1}{\cos^2 \alpha_0} - \frac{\tan \alpha_0 (1-b^2)}{2b} \right] \ln(u-b) + C_H + O(\delta \ln \delta), \quad (88)$$

when $0 < u-b < \delta$. Here C_H denotes such a constant

$$C_H = C_X \tan \alpha_0 - \frac{1}{\cos^2 \alpha_0} \ln \frac{1+b^2}{2} + \frac{1}{\cos^2 \alpha_0} (\ln \cos \alpha_0 - C_\varepsilon). \quad (89)$$

Another estimate of this function is given directly by the expression (67), which involve the integration constant H_B

$$0 < u - b < \delta : \quad H^{(i)}(u) = \frac{(1 - b^4)}{4b(u - b)} + \ln(u - b) + \left[H_B + \frac{3(1 - b^4)}{8b^2} + \ln \frac{2}{b(1 + b)^2} \right] + O(\delta). \quad (90)$$

The estimates (88) and (90) must coincide. Accordingly, in these formulas there must equate the terms with singularity $(u - b)^{-1}$, the terms with logarithmic singularity $\ln(u - b)$ and the terms of order 1. Therefore, we obtain three necessary conditions

$$\frac{(1 + b^2) \tan \alpha_0}{2} = \frac{(1 - b^4)}{4b}, \quad (91)$$

$$\frac{1}{\cos^2 \alpha_0} - \frac{\tan \alpha_0 (1 - b^2)}{2b} = 1, \quad (92)$$

$$C_X \tan \alpha_0 - \frac{1}{\cos^2 \alpha_0} \ln \frac{1 + b^2}{2} + \frac{1}{\cos^2 \alpha_0} (\ln \cos \alpha_0 - C_\varepsilon) = H_B + \frac{3(1 - b^4)}{8b^2} + \ln \frac{2}{b(1 + b)^2}, \quad (93)$$

where the last condition is obtained by using the formula (89).

These three conditions involve trigonometric functions of angle α_0 , which can be expressed through the main auxiliary parameter b using formula (58). As a result, we can verify that the conditions (91) and (92) are satisfied identically

$$\frac{(1 - b^4)}{4b} \equiv \frac{(1 - b^4)}{4b}, \quad \frac{(1 + b^2)^2}{4b^2} - \frac{(1 - b^2)^2}{4b^2} \equiv 1.$$

The last of these three conditions, namely (93), gives the actual relationship for determining the integration constant H_B . Indeed, taking into account (58) and (87), from (93) we obtain

$$H_B = -\frac{(1 - b)(1 + 3b - b^2 + b^3)}{4b^2} + \frac{1 + b^4}{4b^2} \ln \frac{8b^2}{(1 + b^2)^2} + \ln \frac{b(1 + b)^2}{\sqrt{2}(1 + b^2)} - \frac{(1 + b^2)^2}{4b^2} C_\varepsilon.$$

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CONFLICT OF INTEREST

The author of this work declare that they have no conflict of interest.

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